

DUMFRIES DRUNGANS, United KingdomWMO: **031540**Lat: **55.050N**Lon: **3.650W**Elev: **52**StdP: **14.67**Time Zone: **0.00 (EUW)**Period: **94-04**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 22.9	(c) 26.7	(d) 20.5	(e) 15.4	(f) 26.5	(g) 23.2	(h) 17.5	(i) 29.9	(j) 26.7	(k) 48.0	(l) 23.1	(m) 46.9	(n) 0.1	(o) 130	(p) 0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.8	(c) 75.9	(d) 64.9	(e) 71.9	(f) 63.0	(g) 68.6	(h) 61.3	(i) 66.6	(j) 74.1	(k) 64.1	(l) 70.1	(m) 62.1	(n) 67.2	(o) 5.1	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 63.4	(b) 87.5	(c) 70.5	(d) 61.3	(e) 81.1	(f) 67.6	(g) 59.8	(h) 77.0	(i) 65.0	(j) 31.2	(k) 74.3	(l) 29.3	(m) 70.4	(n) 27.9	(o) 67.7	(p) 77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 21.1	(b) 18.4	(c) 15.7	(e) 16.8	(f) 79.4	(g) 6.0	(h) 3.6	(i) 12.5	(j) 82.1	(k) 9.0	(l) 84.2	(m) 5.6	(n) 86.2	(o) 1.3	(p) 88.9		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			16.3	70.7	6.0	3.0	12.0	72.9	8.5	74.6	5.1	76.3	0.7	78.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	39.7	40.7	43.1	46.0	51.8	55.6	59.3	59.8	55.3	49.8	44.6	39.2
	DBStd	8.87	5.41	5.29	4.79	4.77	5.09	4.18	4.27	5.18	3.91	5.42	5.29	6.94
	HDD50	1568	320	262	217	135	40	5	1	1	5	72	174	336
	HDD65	5944	784	681	679	570	410	284	183	176	292	473	613	799
	CDD50	1124	1	1	3	15	94	173	289	306	164	65	11	2
	CDD65	24	0	0	0	0	0	2	6	16	0	0	0	0
	CDH74	192	0	0	0	0	2	21	58	111	0	0	0	0
	CDH80	20	0	0	0	0	0	1	4	15	0	0	0	0
Wind	WSAvg	5.8	6.8	7.9	7.1	5.8	5.7	5.6	4.7	4.2	4.3	6.1	5.5	5.7
Precipitation	PrecAvg	41.3	4.3	3.3	3.1	2.2	2.8	2.5	2.8	3.5	3.9	4.7	4.1	4.2
	PrecMax	54.3	9.1	9.6	6.7	4.7	6.6	4.6	7.7	8.1	8.3	9.8	8.8	8.1
	PrecMin	27.4	0.4	0.1	1.0	0.2	0.3	0.4	0.9	0.5	0.4	0.9	0.7	1.3
	PrecStd	5.4	2.1	2.1	1.5	1.1	1.5	1.1	1.4	1.8	1.9	2.4	2.0	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	53.4	56.2	66.4	72.9	77.4	79.5	81.8	70.0	63.5	58.1	55.5
		MCWB	50.9	50.9	50.0	56.2	64.0	63.5	66.9	68.0	62.7	58.2	56.0	54.1
	2%	DB	50.6	51.6	53.7	59.1	69.0	70.5	74.9	76.8	66.6	60.7	55.8	51.9
		MCWB	48.3	49.3	49.4	53.1	59.9	61.1	64.1	67.0	59.8	57.2	53.4	50.0
	5%	DB	49.2	50.2	51.9	55.7	64.8	66.4	70.4	72.7	64.3	59.3	54.5	50.2
		MCWB	47.0	47.9	48.5	51.0	57.3	58.4	62.1	64.4	58.5	56.4	52.3	48.5
	10%	DB	47.8	48.6	50.4	53.6	60.4	62.9	67.1	68.6	62.3	58.0	52.9	48.6
		MCWB	45.8	46.2	47.1	49.6	54.6	56.3	60.4	62.1	57.8	55.3	50.9	46.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	51.6	52.8	57.6	66.4	65.8	68.6	69.5	63.8	59.4	56.6	54.1
		MCDB	52.7	52.7	53.8	63.1	70.5	74.9	76.6	79.5	68.7	61.4	58.0	55.4
	2%	WB	49.0	49.8	50.7	54.2	61.0	62.0	65.6	67.3	61.4	58.2	54.0	50.4
		MCDB	50.6	51.1	52.3	57.7	67.6	69.2	72.6	75.3	64.8	60.1	55.2	51.6
	5%	WB	47.4	48.4	49.3	52.1	58.0	59.3	62.9	64.9	59.8	57.0	52.5	48.7
		MCDB	48.9	49.9	51.2	54.9	63.5	64.3	68.7	71.5	62.7	59.1	54.1	50.2
	10%	WB	46.0	46.6	47.8	50.4	55.2	57.5	61.0	62.7	58.4	55.5	51.1	46.9
		MCDB	47.7	48.2	49.9	52.9	59.8	61.6	66.0	67.7	61.3	57.9	52.8	48.5
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.2	10.4	12.3	13.3	12.1	12.2	13.8	12.6	11.2	9.1	8.5
		MCDBR	8.6	8.6	11.9	16.4	20.8	19.3	19.7	20.3	16.2	10.9	9.1	9.1
		MCWBR	8.3	7.9	8.8	11.6	13.4	11.7	11.3	11.5	10.5	8.7	8.8	8.9
	5% WB	MCDBR	8.4	7.9	9.8	14.3	19.0	16.4	17.1	18.3	13.4	10.5	8.9	9.2
		MCWBR	8.4	7.9	8.5	11.3	13.7	11.3	11.2	11.5	10.3	9.1	8.9	9.2
Clear-Sky Solar Irradiance	taub	0.328	0.339	0.365	0.384	0.387	0.384	0.385	0.371	0.360	0.344	0.321	0.313	
	taud	2.248	2.336	2.301	2.278	2.317	2.355	2.357	2.406	2.441	2.425	2.331	2.229	
	Ebn at Noon	185	233	254	266	272	273	271	268	255	229	189	161	
	Edh at Noon	19	24	31	37	37	36	36	32	27	22	17	15	
All-Sky Solar Radiation	RadAvg	173	381	725	1167	1517	1499	1445	1169	831	452	222	130	
	RadStd	23	40	69	117	119	149	153	89	69	55	22	12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page